Project Title
A Network Analysis of Candidate–Party Dynamics in Pakistan's National Assembly Elections (2008, 2013, and 2024)

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Course
Social Network Analysis

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Abstract:

This study examines candidate–party dynamics in Pakistan's National Assembly elections (2008, 2013, 2024) using social network analysis of 87 political parties. Constructing a bipartite candidate–party network projected onto a unipartite party-party graph using the Gallup Pakistan Elections Database, the study analyses mobility patterns through network metrics, centrality measures, community detection, and network model comparisons. Findings reveal fragmented integration where low density (7.81%) with full connectivity and small-world properties (low average path length 2.86, high clustering 0.348), indicating strategic rather than random switching. PTI, TLP, and JUI-F emerge as critical brokers with high betweenness centrality. Louvain algorithm detection identifies five ideological and regional clusters with permeable boundaries (low modularity 0.241), where candidates bypass their communities to join central hub parties, suggesting opportunistic rather than ideologically-driven mobility. The network exhibits small-world characteristics. Overall, it indicated weak institutionalisation yet structured elite coordination. This study advances network-based electoral research in South Asia by revealing how candidate circulation shapes Pakistan's evolving multi-party system.

Keywords: Party switching, social network analysis, political networks, electoral dynamics, Pakistan, bipartite networks, centrality measures, community detection, small-world networks

Introduction:

Political parties are central to democratic representation, linking citizens to governing institutions. Studying party-switching behaviour is important because it reveals the stability, coherence, and institutional strength of political systems (Bayer & Malang, 2025). This behaviour has been documented across the United States, Australia, Brazil, and several other countries (Meneghetti et al., 2023). Social network approaches have been very effective in revealing how candidates switch over time, tracing movement within clusters, identifying parties that attract, identifying ideological or strategic motives, and patterns of institutionalisation or fragmentation.

However, in Pakistan, a comprehensive analysis of party-switching behaviour through network approaches is limited. Pakistan's political structure is shaped by party-switching, the rise of independents, and recurrent shifting alliances that impact the institutionalisation. While these patterns are studied, they are not analysed as evolving network structures. Without examining how candidates move between parties across electoral cycles, it remains unclear whether Pakistan's party system is consolidating or fragmenting, and which parties hold central positions. This study aims to reveal these patterns by focusing on three electoral cycles: 2008, 2013, and 2024 National Assembly elections. The research objectives are:

1. How stable are candidate–party affiliations across the three elections, and what do these patterns show about loyalty, fragmentation, and candidate circulation?
2. Which political parties attract the highest number of switching candidates, and how does this pattern reflect their centrality, influence, or brokerage position within Pakistan's political network?
3. How has the overall structure of the candidate–party network evolved, and what does this indicate about the institutionalization or fragmentation of Pakistan's party system and the nature of party-switching?

The paper begins with a review of global and local literature on party switching, followed by sections on data, preprocessing, and network construction. It then presents network metrics, visualisations, and empirical findings, discusses their implications for Pakistan's political system, and concludes by highlighting the study's contributions and future research pathways.

Literature Review:

Social Network Analysis (SNA) is being increasingly employed to study electoral politics. In the context of party-switching networks, by mapping ties between candidates and parties, SNA helps identify clusters and highlights which parties consistently attract the most candidates. This review synthesises studies utilising a network-based approach to understand electoral politics, particularly party switching and contextualises it within Pakistan's electoral politics, emphasising the need for social network analysis.

Bayer & Malang (2025) and Faustino et al. (2019) use advanced network modelling in party-switching contexts. Bayer & Malang employ Exponential Random Graph Models (ERGM) with Markov Chain Monte Carlo Maximum Likelihood Estimation (MCMC MLE) to analyse the impact of party switching on parliamentary behaviour in Zambia, demonstrating how MPs realign network ties after switching parties while maintaining former connections. Faustino et al. utilise Stochastic Block Models (SBM) combined with beta distribution modelling to analyse ideological clustering in Brazilian party-switching networks, revealing that switching follows ideological rather than opportunistic patterns. Building on these approaches, this study adopts Faustino et al.'s method of constructing networks where nodes represent parties and edges reflect the number of politicians switching between them, applied across the 2008, 2013, and 2024 electoral cycles. Similarly, Brito et al. (2020) use complex network theory and the Leiden algorithm, which is an improved version of the Louvain algorithm for community detection, to analyse voting pattern similarities in Brazil's legislature, demonstrating how network analysis reveals evolving political coalitions across legislative periods. Inspired by their approach, this study uses the Louvain algorithms to identify clusters and assess whether switching behaviour is ideological or strategic.

Despite these methodologies, gaps remain. Existing studies mostly focus on detecting clusters or modelling switching probabilities, and giving limited attention to centrality measures. Thus, questions like which parties act as brokers, bridges, or attractors of political mobility are underexplored.

Furthermore, there is limited use of SNA in electoral politics in the South Asian context, particularly in Pakistan. Existing studies in Pakistan mainly use statistical methods. Zhirnov & Mufti (2019) employ logistic regression models, revealing that switching has immensely increased after the 2000s and that candidates with personal votes find it easier to switch parties, indicating weaker party-voter linkages. Similarly, Wu & Ali (2020) document Pakistan's structural transformation from a two-party to a three-party system with PTI's emergence during 2013-2018, emphasising how this shift impacted electoral dynamics and party-switching. However, neither study employs network analytical methods, failing to capture structural patterns, clustering tendencies, or identify parties that are central to this party-switching behaviour.

Addressing these gaps, our study expands the scope of existing research by constructing a bipartite candidate–party network across three electoral cycles of Pakistan and then projects it into a unipartite party–party graph. It further includes analysing centrality metrics with community detection algorithms to identify parties that function as bridges and those that consistently attract candidates, and identify movement across clusters.. Through this, this study aims to contribute to understanding the political structure, process of institutionalisation and fragmentation in Pakistan.

Data and Network Construction:

The dataset was sourced from the Gallup Pakistan Elections Database, containing 24,585 electoral records spanning 1970-2024 for National Assembly elections. We extracted data in CSV format, filtering for three election years: 2008, 2013, and 2024. The 2018 election was excluded due to missing candidate names across all 3,353 records. Data preprocessing involved systematic standardisation steps. We consolidated approximately 280 unique party names to 250 by resolving naming variations and implemented an abbreviation system for over 80 parties with lengthy names (e.g., "Muttahidda Majlis-e-Amal Pakistan" to "MMA"), preserving originals for verification. Candidate names were standardised through title-case formatting and whitespace normalisation. A critical preprocessing decision involved removing all independent candidates from the dataset, which constituted approximately 4,000-5,000 records. This exclusion was justified by our research focus on party-based political structures rather than individual candidacies, which improved the analytical lens on inter-party dynamics and formal organisational networks. We constructed a bipartite network where nodes represent political parties and candidates, with edges connecting candidates to their affiliated parties. Edge weights show the number of electoral cycles a candidate contested under each affiliation. The network was subsequently projected onto a unipartite party-party graph, where nodes represent only parties and edges signify shared candidate pools between parties, weighted by the intensity of candidate-sharing relationships across the three election cycles.

Methodology:

This study employs comprehensive network analysis techniques combining local and global metrics to analyse the connectivity and movement of candidates within the structure. We calculated different centrality measures to identify influential parties from different structural perspectives. Degree centrality captures the number of direct connections each party maintains, indicating the breadth of candidate-sharing relationships. Betweenness centrality measures how often a party lies on the shortest paths between other parties, suggesting brokerage roles within the political structure. Eigenvector centrality accounts for the quality of connections by assigning higher scores to parties connected to other well-connected parties. Additionally, we examined the clustering coefficient to assess the tendency of parties to form tightly-knit triangular relationships, and analysed degree distribution patterns to test for scale-free network properties. For community detection, we applied the Louvain algorithm, a modularity-optimisation method that iteratively aggregates nodes to maximise within-group connections relative to between-group connections, identifying naturally occurring party clusters sharing candidate pools or electoral strategies. We validated structural findings by comparing the empirical network against three theoretical models: Erdős-Rényi random graphs, Watts-Strogatz small-world networks, and Barabási-Albert scale-free networks. Visualisation and computation utilised the igraph package in R, generating network layouts, statistical summaries, and graphical representations of centrality distributions, community structures, and degree patterns for robust characterisation of Pakistan's party network topology.

Analysis and Discussion:

Global Level of Analysis:

Metric	Value
Number of Parties (Nodes)	**87**
Number of Connections (Edges)	**292**
Network Density	**0.0781 (7.81%)**
Average Path Length	**2.86 steps**
Network Diameter	**6 steps**
Global Clustering Coefficient	**0.348**
Average Local Clustering Coefficient	**0.674**
Number of Connected Components	**1**
Largest Component Size	**87 parties**
Largest Component Percentage	**100.0%**

Figure 1: Overview of network metrics

As highlighted in Figure 1, this network has a low density (7.81%), indicating a sparse and weak cohesive structure, reflecting institutional fragmentation. Despite this, the average path length (2.86) is very low, indicating a closely integrated party system where any two parties are connected through fewer than three intermediary parties. Diameter of 6 steps represents the longest chain of candidate movements. Both metrics and high local clustering coefficient demonstrate small-world properties, suggesting that Pakistan's fragmented multi-party system is structurally integrated through candidate circulation. The presence of a single component indicates that every party, including small regional groups, is reachable through switching chains, demonstrating full system-wide integration despite the presence of 87 parties.

Collectively, these metrics indicate a structural paradox where low density (7.81%) is combined with full connectivity (1 component, 100%) and high local clustering (0.674). This means candidate mobility is not random but instead flows through selective, structured pathways. The pattern aligns with Zhirnov and Mufti's (2019) argument that switching increased significantly after the 2000s, helping explain the network's extensive pathways and small-world characteristics. Low cohesion based on density shows a structure with weak institutionalisation, while the non-random circulation of elites reflects a strategically coordinated environment. This interpretation is supported by studies, which show that the system is institutionally weak and fragmented, yet strategically coordinated through elite mobility networks (Wu & Ali, 2020; Zhirnov & Mufti, 2019).

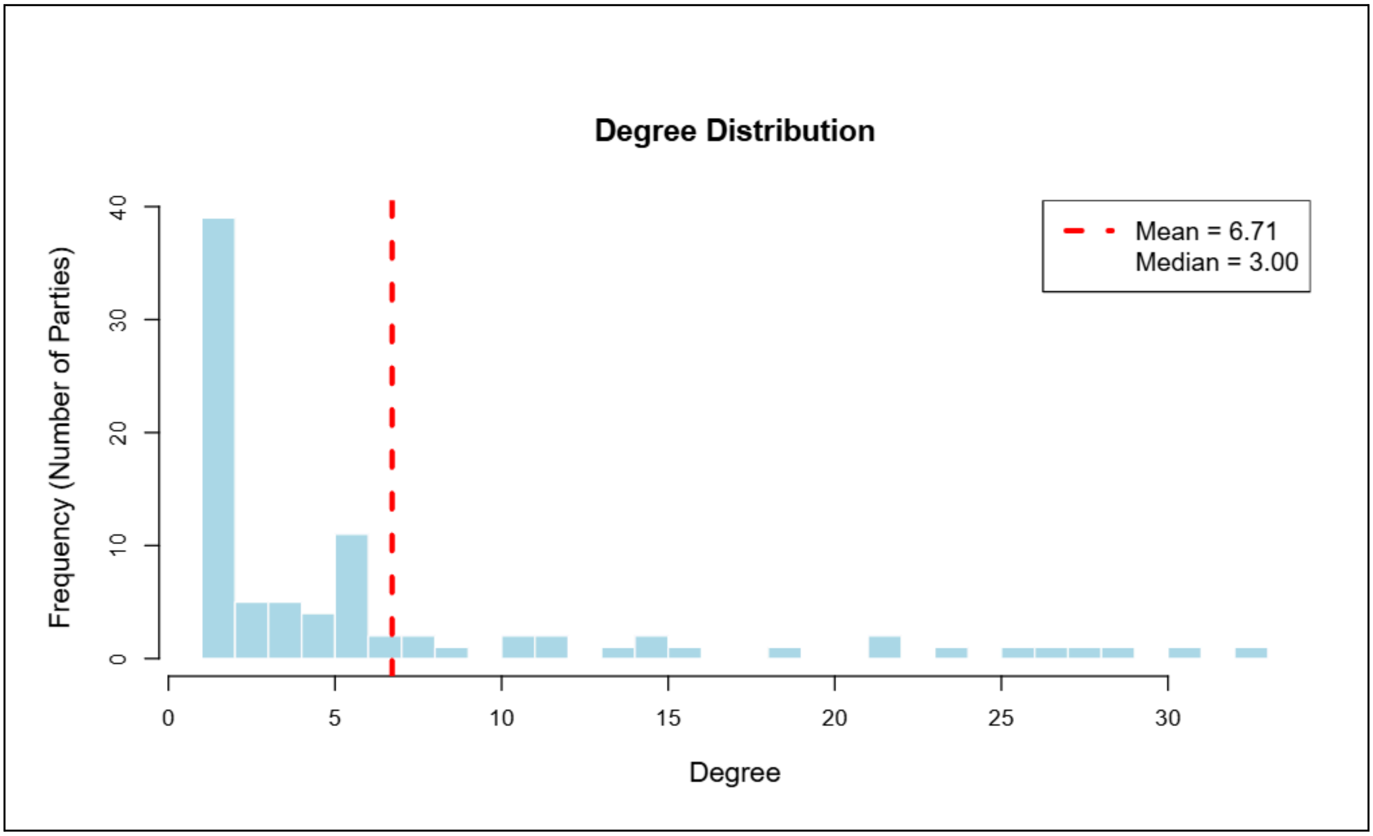

Figure 2: Degree distribution (histogram)

Figure 2 reveals a highly skewed distribution, with most parties holding very few connections. The gap between the Mean Degree (6.71) and Median Degree (3.00) shows strong centralisation, where a small group of high-degree parties drives most switching. This suggests candidate shifting is not broadly democratic but concentrated in a core of dominant hubs. This is consistent with literature showing how there are major parties (Hubs) that dominate candidate mobility, giving a core-periphery structure (Wu & Ali, 2020; Zhirnov & Mufti, 2019).

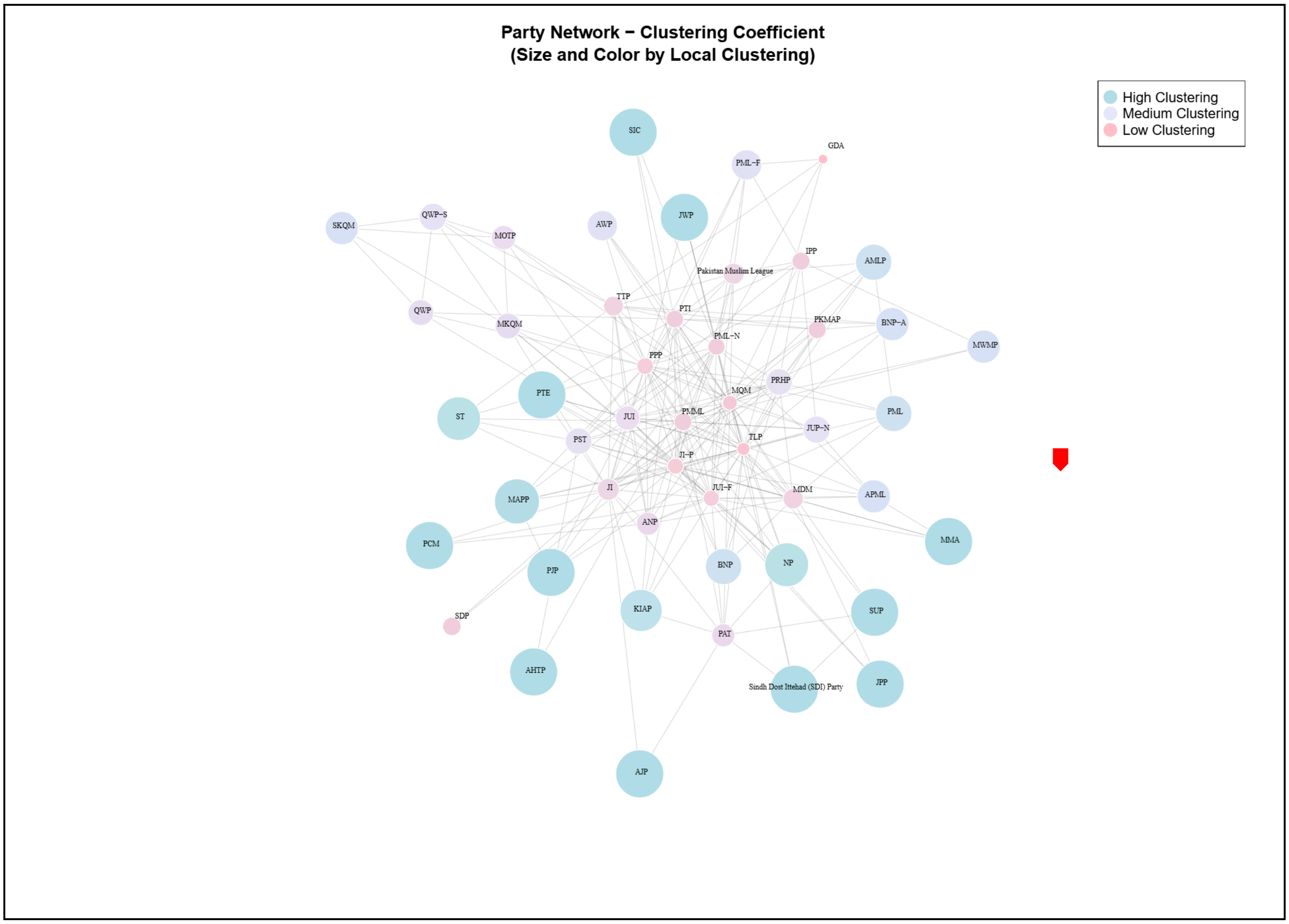

Figure 3: Local Clustering Coefficient

There is a stark difference between global clustering (0.348) and local clustering (0.674) shows that while parties form dense internal communities, overall cohesion is moderate. Many peripheral (ST, MMA, AJP) have a high clustering coefficient, operating as tightly knit factions where candidate movement is primarily confined to proximate groups. In contrast, major parties such as PTI, PPP, PML variants, JI-P, and TLP (pink/light purple nodes) have medium to low clustering. This structural pattern confirms that these major parties function as generalised network sinks, having candidates from diverse and unconnected political blocs across the system. This presents an opportunistic alignment going beyond ideological motivations that persist in the periphery.

Centrality Analysis:

Degree	Betweenness	Eigenvector
TLP (33)	TLP (672.728)	PPP (1.0)
MQM (31)	MQM (517.172)	MQM (0.993)
PPP (29)	PTI (448.46)	TLP (0.978)
JI-P (28)	JUI-F (442.23)	PMML (0.973)
PMML (27)	JI-P (419.478)	JI-P (0.94)

Figure 4: Centrality measures by Top 5

TLP's high degree centrality (33 connections) and high betweenness are unexpected, considering its recent emergence. Literature shows a pattern in Pakistan where high switching rates and weak institutionalisation allow new parties to rapidly establish extensive connections by attracting candidates from other parties (Zhirnov & Mufti, 2019). Similarly, PTI's high betweenness demonstrated its broker role, linking regional, religious, and mainstream parties. This supports Wu & Ali's (2020) argument that PTI's rise restructured Pakistan's party system by attracting candidates across traditional ideological boundaries, transforming from an outsider into a key connector. Zhirnov & Mufti's (2019) observation that candidates with personal votes switch more easily is evident here as parties with high betweenness (PTI, JUI-F) likely attract locally influential candidates capable of moving across party boundaries. The study also specifically discusses how PTI tend to have candidates who had strong local bases of support and records for winning elections from other political parties. Thus, shows PTI became a hub precisely by accepting candidates across traditional boundaries.

Eigenvector centrality reveals PPP as a central node, as it is connected with other influential parties through candidate switching, reflecting PPP's historical role as a major mainstream party. These findings are consistent with both studies showing PPP as the major party. Wu & Ali (2020) also identify PPP as having a strong influence on the overall structure historically and having electoral consistency. Similarly, MQM's high eigenvector centrality (0.993) indicates its importance within urban Sindh's political network.

Community Detection:

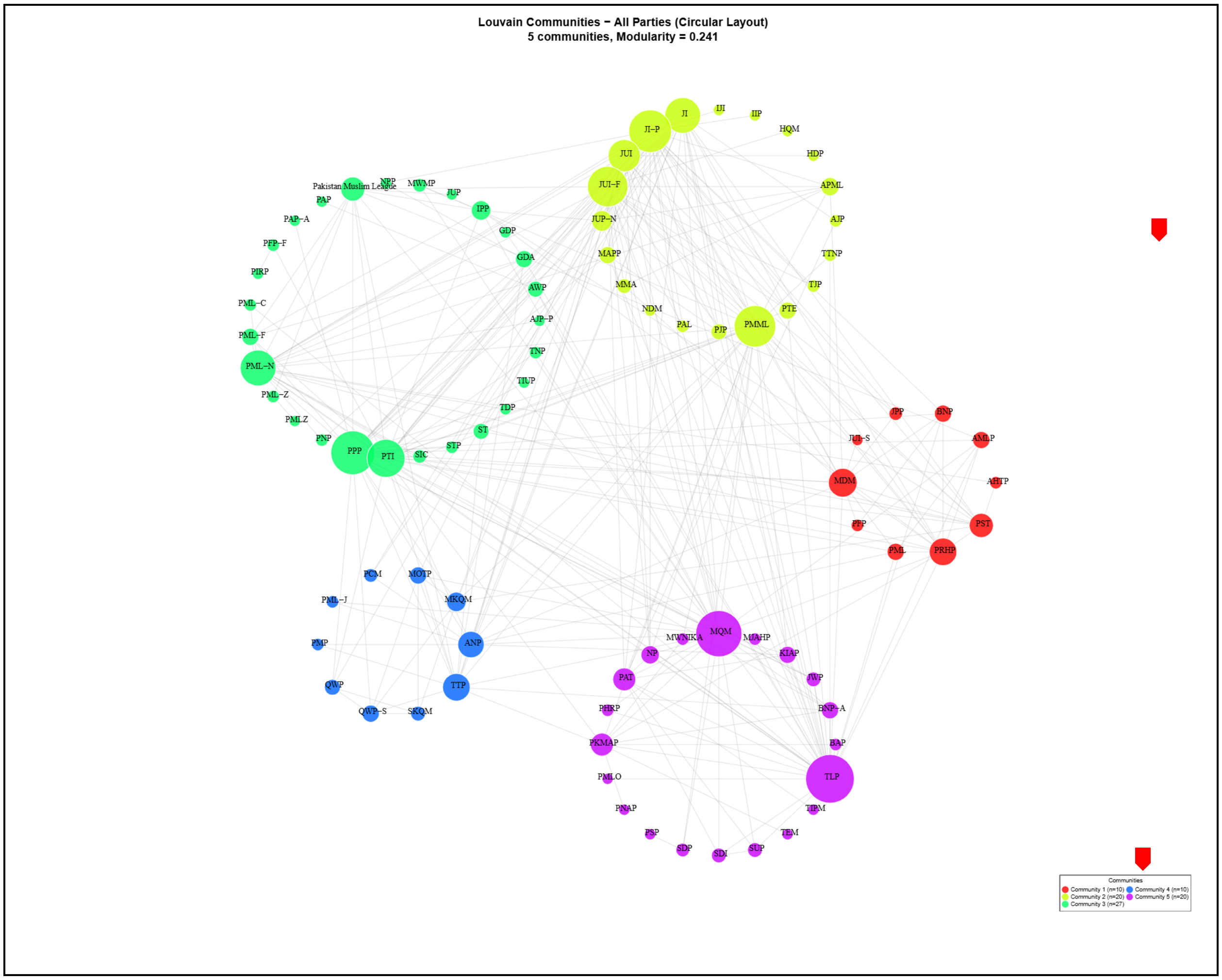

Figure 6: Louvain Community Algorithm

The Louvain algorithm reveals five distinct communities, modularity = 0.241, representing ideological, regional, and strategic divisions in Pakistan's party system. Green Cluster includes mainstream parties (PPP, PTI, PML-N) forming the largest hub, attracting switchers from all communities. Yellow Cluster includes religious parties (JUI-F, JI-P, JUP-N, MMA) that show ideological coherence. Red Cluster includes regional/ethnic parties (PST, PKMAP, ANP). Blue Cluster includes regional and small mainstream variants (ANP, PML-J). Purple Cluster includes MQM (urban Sindhi/Muhajir), TLP (religious hardliner), BNP-A (Baloch nationalist), and BAP (Balochistan regional) represent distinct ethnic, religious, and regional identities. Their co-clustering suggests opportunistic candidate circulation rather than ideological, thus the switch among them is based on strategic positioning. Similarly, low modularity also shows that the structural boundaries between them are permeable and extensive cross-community mobility. The extensive inter-cluster edges confirm that candidates bypass ideological boundaries to join strategic hubs, showing a trend of opportunistic mobility driven by power concentration rather than ideological consistency. This pattern was also observed in analysis of local clustering coefficient.

These patterns are consistent with dynamics described in the literature. Zhirnov & Mufti (2019) highlights how weak party-voter linkages, weak institutionalisation, and high volatility enable opportunistic switching. Wu & Ali (2020) show PTI combining across ideology with elite recruitments, Green Clusters show this as PTI is part of the mainstream and diverse community. While there exists some ideological clustering (high local clustering in peripheral parties as shown earlier), but dominant pattern is of opportunistic cross-boundary switching. With 87 parties organised into five communities yet connected through switching pathways, the network demonstrates a fragmented but strategically integrated system. Both studies show that this is due to elite circulation, not due to institutional development.

Comparison with seminal network models:

Metric	Party_Party	Random_ER	SmallWorld_WS	ScaleFree_BA
Average Path Length	2.862	2.547	3.073	2.531
Avg Clustering Coefficient	0.348	0.064	0.314	0.11
Power Law Coefficient	0.914	0.081	2.561	1.536

Figure 7: Network Model Comparison

The party-switching network has a low average path length (2.862) that is similar to the Scale Free Network (2.531)and Random Network (2.547), and lower than the Small World Network (3.073), indicating efficient party switching where parties connect through short switching chains. The power-law exponent (0.914) is less than the Scale-Free network (1.536) and exceeds the Random Network (0.081), confirming the presence of hubs. Few parties disproportionately attract candidates, acting as central mobility brokers. The network high clustering coefficient (0.348) is higher than Random Network, indicating that the network is more structured, reflecting properties of real-world networks. Overall, the high clustering and low average path length depict a small-world, real-life network with the presence of a hubs.

Limitations:

This study offers insights into party switching but has key constraints. It does not account for independent candidates or use predictive models like ERGM or SBM to test opportunistic versus ideological motives. Temporal detail is missing in the unipartite network, limiting the detection of specific switching periods or intra-election moves. Contextual factors like electoral performance, candidate traits, and external influences are not analysed. Future studies can work on these gaps.

Conclusion:

This study applied social network analysis to Pakistan's candidate–party dynamics across the 2008, 2013, and 2024 elections, revealing a system of fragmented integration. Despite 87 parties and low network density (7.81%), full connectivity through strategic switching pathways indicated a structured elite coordination rather than random. Degree distribution highlighted the presence of hub parties. Centrality analysis identified PTI and TLP as critical brokers linking otherwise separate ideological and regional parties and validating PTI's transformative role in reshaping Pakistan's party system. Louvain algorithm community detection revealed five distinct clusters with permeable boundaries (low modularity 0.241). Extensive cross-boundary switching, where candidates bypass ideological communities to join central hub parties, suggests mobility is driven by opportunistic motives and strategic positioning, consistent with literature findings that candidates with personal votes switch more easily. The findings reflected weak institutionalisation yet strategically coordinated through elite mobility. Network comparison confirmed small-world properties with a high clustering coefficient and a short average path length, suggesting efficient candidate mobility through tightly-knit local communities connected by strategic bridges. By constructing the first multi-cycle party-switching network for Pakistan, this study contributes a structural understanding of political mobility, highlighting a party system that is organisationally weak yet strategically interconnected.

References

Bayer, A., & Malang, T. (2025). Late to the party: Do party switchers behave differently in parliament from party loyalists? *International Political Science Review.* https://doi.org/10.1177/01925121251343841

Brito, A. C. M., Silva, F. N., & Amancio, D. R. (2020). A complex network approach to political analysis: Application to the Brazilian Chamber of Deputies. *PLoS ONE, 15*(3), e0229928. https://doi.org/10.1371/journal.pone.0229928

Faustino, J., Barbosa, H., Ribeiro, E., & Menezes, R. (2019). A data-driven network approach for characterisation of political parties' ideology dynamics. *Applied Network Science, 4*, 48. https://doi.org/10.1007/s41109-019-0161-0

Meneghetti, N., Pacini, F., Biondi Dal Monte, F., Cracchiolo, M., Rossi, E., Mazzoni, A., & Micera, S. (2023). Predicting party switching through machine learning and open data. *iScience, 6*(10), 107098. https://doi.org/10.1016/j.isci.2023.107098

Wu, X., & Ali, S. (2020). The novel changes in Pakistan's party politics: Analysis of causes and impacts. *Chinese Political Science Review, 5*, 513–533. https://doi.org/10.1007/s41111-020-00156-z

Zhirnov, A., & Mufti, M. (2019). Electoral constraints on inter-party mobility of candidates. Comparative Politics*, 51*(4), 519–537. https://www.jstor.org/stable/26663946